$$p(x) = \int p(x,z)\,dz$$

$$\log p(x) = \log \int p(x,z)\,dz$$

$$= \log \int p(x,z) \cdot \frac{q_\phi(z|x)}{q_\phi(z|x)}\,dz.$$

$$\int p(x) \log \frac{p(x)}{q(x)}\,dx = D_{KL}(p\|q)$$

$$= E_{x\sim p}\left[\log \frac{p(x)}{q(x)}\right]$$

$$= \log E_{z\sim q_\phi}\left[\frac{p(x,z)}{q_\phi(z|x)}\right]$$

$$\geq E_{z\sim q_\phi}\left[\log \frac{p(x,z)}{q_\phi(z|x)}\right]$$

log 특성 Concave 성질

$$= E_{z\sim q_\phi}\left[\log \frac{p(x|z)\,p(z)}{q_\phi(z|x)}\right]$$

$$= E_{z\sim q_\phi}[\log p(x|z)] - E_{z\sim q_\phi}\left[\log \frac{q_\phi(z|x)}{p(z)}\right]$$

$$= E_{z\sim q_\phi}[\log p(x|z)] - D_{KL}(q_\phi(z|x)\,\|\,p(z))$$

Reconstruction Matching Term

$$\mathcal{L} = -E_{z\sim q_\phi}[\log p(x|z)] + D_{KL}(q_\phi(z|x)\,\|\,p(z))$$

Reconstruction Matching Term

$$p_\theta(x|z) = \mathcal{N}(x;\mu_\theta, I)$$

$$q_\phi(z|x) = \mathcal{N}(z;\mu_\phi,\sigma_\phi^2 I) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \qquad p(z) = \mathcal{N}(0, I)$$

① Matching Term

$$D_{KL}(q_\phi(z|x)\,\|\,p(z))$$

$$= E_{z\sim q_\phi}[\log q_\phi(z|x) - \log p(z)]$$

$$= E_{z\sim q_\phi}[\log q_\phi(z|x)] - E_{z\sim q_\phi}[\log p(z)]$$

Note that $\mathcal{N}(\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$

$$\rightarrow \left(\log \frac{1}{\sqrt{2\pi}\sigma} + E\left[-\frac{(z-\mu)^2}{2\sigma^2}\right]\right) - \left(\log \frac{1}{\sqrt{2\pi}} + E\left[-\frac{z^2}{2}\right]\right)$$

Note $E[(z-\mu)^2] = \sigma^2$, $E[z^2] = Var(z) + E[z]^2 = \sigma^2 + \mu^2$

$$\rightarrow \log \frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{2} - \log \frac{1}{\sqrt{2\pi}} + \frac{(\sigma^2+\mu^2)}{2}$$

$$= -\log\sigma - \frac{1}{2} + \frac{1}{2}(\sigma^2+\mu^2)$$

$$= -\frac{1}{2}(1 + \log\sigma^2 - \sigma^2 - \mu^2)$$

$$\mathcal{L} = -\mathbb{E}_{z\sim q_\phi}[\log p(x|z)] + D_{KL}(q_\phi(z|x) \| p(z))$$

Reconstruction — Matching Term

$$p_\theta(x|z) = \mathcal{N}(x; \mu_\theta, \sigma^2 I)$$

$$q_\phi(z|x) = \mathcal{N}(z; \mu_\phi, \sigma_\phi^2 I) \qquad p(z) = \mathcal{N}(0, I)$$

① **Matching Term**

$$D_{KL}(q_\phi(z|x) \| p(z))$$

$$= \mathbb{E}_{z\sim q_\phi}[\log q_\phi(z|x) - \log p(z)]$$

$$= \mathbb{E}_{z\sim q_\phi}[\log q_\phi(z|x)] - \mathbb{E}_{z\sim q_\phi}[\log p(z)]$$

note that $\mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$

$$\rightarrow \left(\log \frac{1}{\sqrt{2\pi}\sigma} + \mathbb{E}\left[-\frac{(z-\mu)^2}{2\sigma^2}\right]\right) - \left(\log \frac{1}{\sqrt{2\pi}} + \mathbb{E}\left[-\frac{z^2}{2}\right]\right)$$

note $\mathbb{E}[(z-\mu)^2] = \sigma^2$, $\mathbb{E}[z^2] = Var(z) + \mathbb{E}[z]^2 = \sigma^2 + \mu^2$

$$\rightarrow \log \frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{2} - \log \frac{1}{\sqrt{2\pi}} + \frac{(\sigma^2+\mu^2)}{2}$$

$$= -\log \sigma - \frac{1}{2} + \frac{1}{2}(\sigma^2 + \mu^2)$$

$$= -\frac{1}{2}(1 + \log \sigma^2 - \sigma^2 - \mu^2)$$

② **Reconstruction Term**

$$\mathbb{E}_{z\sim q_\phi}[\log p_\theta(x|z)]$$

Monte Carlo

$$\simeq \frac{1}{N} \sum \log p_\theta(x|z)$$

$$= MSE$$

1. $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \sigma_\phi^2(x) I)$ → does not depend on integral target 'z'!

Thus, $D_{KL}(q_\phi(z|x) \| p(z)) = \int q_\phi(z|x) \log \frac{q_\phi(z|x)}{p(z)} dz$ is tractable

2. $p_\theta(x|z) = \mathcal{N}(\mu_\theta(z), \sigma^2 I)$ → does depend on 'z'

Thus, $\mathbb{E}_{z\sim q_\phi}[\log p_\theta(x|z)] = \int q_\phi(z|x) \cdot \log p_\theta(x|z)\, dz$ is intractable